

√1

$$\int \sin x \cos x dx$$

- a) substitution $u = \sin x$
- b) substitution $u = \cos x$
- c) use $\sin 2x = 2 \sin x \cdot \cos x$
- d) integral by parts

√2

$$a) \int \frac{1}{x^2} \cos\left(\frac{1}{x}\right) dx$$

$$b) \int \frac{x^3}{1+x^4} dx$$

$$c) \int x^3 \sqrt{x} dx$$

$$d) \int_0^1 \frac{1}{\sqrt{x}} dx$$

√2

a) $\int \frac{1}{x^2} \cos\left(\frac{1}{x}-1\right) dx$

b) $\int \frac{x^3}{1+x^4} dx$

c) $\int x^3 \sqrt{x^2+1} dx$

d) $\int_0^1 \frac{x}{\sqrt{1-x^4}} dx$

√3

Determine the minimal of $a > 0$ such that the area of the region limited by

$y = \sin^2 x, y = 0, x = 0$ and $x = a$ is equal to the area of the region

limited by $y = \cos^2 x, y = 0, x = 0$ and $x = a$

√4

Find all positive and differentiable f on $(-\frac{1}{3}; \infty)$ such that

$(3x+1)^2 f(x) f'(x) = f^2(x) : f(0) = 6e^{-1/3}$